

Briefing · May 2026 · Iceland

MicroPulse × Catapult IMA

Operationalised inertial sport science for football

Catapult IMA is the most under-utilised data layer in the platform. Most clubs export Bands 5-8 stride counts, IMA CoD L/R, and IMA decel events but never translate them into a coach decision. MicroPulse is the analytics layer that closes that loop. Every IMA-derived metric in MicroPulse is tied to a published sport-science paper, surfaced as a clear coach action, and (for ELITE teams) refined by a Claude-Haiku layer that reads free-text wellness notes.

This document is a 1-page summary of the six IMA-derived modules we have shipped and the coach decisions each one drives. It is designed to be shareable inside Catapult's S&C curriculum.

Catapult IMA source	MicroPulse module	Coach decision surfaced	Sport-science citation
Free Running Bands 5-8 (high-velocity stride counts)	Sprint Exposure	Weekly bands 5-8 sum vs match-day demand. Under 50% → 3× hamstring injury risk → coach adds sprint volume. Over 150% → spike → coach caps.	Malone 2018 (BJSM)
Free Running 8-band stride rate + GPS distance	Stride Intelligence	Cadence + stride length + L/R asymmetry. Coach sees if today's mechanics match personal baseline; flag when stride length drops > 10% (Buchheit 2014 norms).	Buchheit 2014
IMA CoD L/R high-tier event counts (14d)	L/R CoD Asymmetry	≥ 15% asymmetry → 3× non-contact lower-limb injury risk. Triggers automatic swap to unilateral B-stance RDL on weaker side in next strength session.	Bishop 2020
IMA Band 3 decel event counts	McBurnie Decel Intelligence	Preferred decel source (better indoor + outdoor coverage than GPS Gen-2 B3+). Drives overload / underload / coupling / concentration verdicts. Auto-strips Nordic if 3+ consecutive HIGH-burden days.	McBurnie 2022
IMA Free Running + Metabolic Power (no GPS)	Indoor Composite Load	IMU-only weighting (HMLD and Decel B2-3 explicitly dropped) for indoor sessions. Sport-agnostic — handball + basketball + futsal teams without GPS vests still get a load score.	Brown 2016, Osgnach 2023
Free Running Bands 1-8 stride counts (all bands)	Stride length formula (v3)	Total distance ÷ total strides (session avg). Stable rest-day baseline. Detects mechanical compensation patterns when no high-velocity work happens.	Buchheit 2014

Live demo flow — one player, one match-week

What the meeting actually looks like in 3 minutes. Pick any player who has IMA Free Running data and play these screens in order:

- **1.** Open Decision Summary card. Coach sees Sprint Exposure chip showing 42% of match demand (UNDERLOAD band, Malone 2018). Engine prescribes: add sprint block at MD-3.
- **2.** Open Stride Intelligence card. Cadence 87 strides/min, stride length 1.78 m, L/R asymmetry 18% (left weaker). Bishop 2020 chip lights up.
- **3.** Open Strength card. MD-4 microdose: Trap-bar DL 3×3 cluster, Nordic 2×5, Copenhagen 2×6 — plus B-stance RDL added on weaker leg (CoD asymmetry rule fired automatically).
- **4.** Click 'AI refinement'. Player wrote 'Mjóhryggur stífur eftir leik' in their wellness check-in this morning. Claude (ELITE-only) suggests swapping the Trap-bar DL to Front squat — lower spinal compression while keeping the strength stimulus. Coach approves with one click.
- **5.** Click 'Send to player's app'. Player receives push notification + structured session in their PWA. Same flow works for the whole squad (one click sends to all active players).

Five clicks. Five sport-science papers operationalised. Every step traceable to a citation and a coach decision. This is what Catapult IMA looks like when it is treated as a decision layer, not a dashboard.

Why this matters for Catapult education and curriculum

- **Closes the activation gap:** Most coaches use 15-20% of Catapult's IMA capability. Bands 5-8 sit untouched, IMA CoD is exported but rarely acted on, IMA Band 3 is buried in OpenField. MicroPulse turns these into one-line coach decisions.
- **Teachable example:** Every IMA module in MicroPulse cites a published paper. A Catapult educator can point at MicroPulse and say 'this is what Malone 2018 looks like in practice'.
- **Premium-Catapult-friendly:** We support Premium-IMU plans (Bands 5-8, CoD L/R, IMA Band 3) as the primary data source. We also have a Lite-tier path (Foster + RPE-based monitoring) for clubs without the Premium IMU contract — extending Catapult's value proposition into lower divisions where they currently lose deals to cheaper monitoring tools.
- **Production traction:** Iceland is the testbed. 13 Catapult-using clubs, 4 already on MicroPulse (Breiðablik live, Þór ELITE, Grindavík onboarding, Afturelding Lite). Operational results, not slides.

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Available for a deeper walkthrough — typical IMA-stack demo runs ~15 minutes.